ECE 750T — Midterm Reference Sheet
(Nothing you write on this page will be graded)

- *Binary outcomes.* Let $Y \in \{0,1\}$ denote the observed outcome, $\widehat{Y} \in \{0,1\}$ denote a prediction or decision, and $A \in \{a,b\}$ denote a group attribute (i.e. there are two disjoint groups, a and b). Without loss of generality, we use $Y = 1$ to represent the positive observed outcome (e.g., performs well on the job) and $\widehat{Y} = 1$ to represent a positive prediction or decision (e.g., receives a job offer).

	$Y=1$	$Y=0$
$\widehat{Y}=1$	TP	FP
$\widehat{Y}=0$	FN	TN

  The true-positive rate (TPR), false-positive rate (FPR), and positive predictive value (PPV) are

$$\begin{aligned}\text{TPR} &= \frac{TP}{TP+FN} = P(\widehat{Y}=1 \mid Y=1),\\ \text{FPR} &= \frac{FP}{FP+TN} = P(\widehat{Y}=1 \mid Y=0),\\ \text{PPV} &= \frac{TP}{TP+FP} = P(Y=1 \mid \widehat{Y}=1).\end{aligned}$$

  Compute TPR_a as the TPR on group a data only, TPR_b as the TPR on group b data, and so on. For group a, the positive-decision rate is $P(\widehat{Y}=1 \mid A=a)$ and prevalence is $P(Y=1 \mid A=a)$.

$$P(\widehat{Y}=1 \mid A=a) = \text{TPR}_a P(Y=1 \mid A=a) + \text{FPR}_a P(Y=0 \mid A=a).$$

- *Statistical group criteria.* The following conditional comparisons are defined only when their conditioning events have positive probability:
  - *Demographic parity (independence):* equal positive-decision rates.
$$P(\widehat{Y}=1 \mid A=a) = P(\widehat{Y}=1 \mid A=b).$$
  - *Equal opportunity:* equal true-positive rates.
$$P(\widehat{Y}=1 \mid Y=1, A=a) = P(\widehat{Y}=1 \mid Y=1, A=b).$$
  - *Equalized odds (separation):* equal TPR and FPR, or equal positive-decision rates conditional on each outcome $Y=y$.
$$P(\widehat{Y}=1 \mid Y=y, A=a) = P(\widehat{Y}=1 \mid Y=y, A=b), \quad y=0,1.$$
  - *Group calibration:* for a probability-valued score R, each group a and supported score r satisfies
$$P(Y=1 \mid R=r, A=a) = r.$$

- *k-anonymity.* For the quasi-identifier columns specified in a question, a released table is k-anonymous if every combination of values that occurs in the table appears in at least k rows.

- *Neighboring datasets.* Two datasets $D \sim D'$ are neighbors if they differ only in one unit's contribution, i.e. D' can be derived from D by adding or removing a single row.

- *Differential privacy.* A randomized mechanism M is (ϵ, δ)-DP if, for every ordered pair of neighboring datasets $D \sim D'$ and every event S of possible outputs,

$$\Pr[M(D) \in S] \le e^{\epsilon} \Pr[M(D') \in S] + \delta.$$

  A mechanism is ϵ-DP (also called "pure" DP) if it satisfies the above requirement with $\delta = 0$.

- *Composition and post-processing.* If there are T differentially private mechanisms, where the t-th mechanism is (ϵ_t, δ_t)-DP, then a new mechanism that composes all of them is

$$\left(\sum_{t=1}^{T} \epsilon_t, \sum_{t=1}^{T} \delta_t\right)\text{-DP}.$$

  Processing a released DP output without additional private-data access preserves its guarantee.